A2	PROSOIL	BH NO: 266+240	Longitude: N 25^{0}05.737$'$	Latitude: E 92^{0}06.773$'$
		Location: Sarighat, Nodirpar, Goainghat, Sylhet.		CHAINAGE: 266+240
Name of Project:Geotechnical Investigation for Subregional Transport Project Preparatory Facility.		Type of Drill Rig: Manual Rotary	Borehole Depth: 42.0m	RL: -14.7
CLIENT : Hifab International.		Commencement Date: 15-09-2014	Completed Date: 18-09-2014	G. W. T : 30.0m above from E.G.L.

R.L./ THICKNESS	SOIL STRATA	BORE LOG	SAMPLING From (m)	SAMPLING To (m)	SPT-N Value Blow/300mm N1=150 (mm)	N2=300 (mm)	N3=450 (mm)	N=N2+N3	DISTURBED SAMPLE	DEPTH (m) SPT VALUES (BLOWS / 300mm)	% Sand	% Passing 200	% LL	% PI	SP Gravity	% W_n
-15 to -66	0 to 51		1.5	1.95	1	2	2	4	D-1	4	1	2	3	4	5	6
			3.0	3.45	1	1	2	3	D-2	3	7	8	9	10	11	12
			4.5	4.95	1	2	1	3	D-3	3	13	14	15	16	17	18
			6.0	6.45	4	4	5	9	D-4	9	19	20	21	22	23	24
			7.5	7.95	5	7	8	15	D-5	15	25	26	27	28	29	30
			9.0	9.45	5	10	12	22	D-6	22	31	32	33	34	35	36
			10.5	10.95	5	8	10	18	D-7	18	37	38	39	40	41	42
			12.0	12.45	4	9	10	19	D-8	19	43	44	45	46	47	48
			13.5	13.95	3	9	8	17	D-9	17	49	50	51	52	53	54
			15.0	15.45	7	8	6	14	D-10	14	55	56	57	58	59	60
			16.5	16.95	8	6	13	19	D-11	19	61	62	63	64	65	66
			18.0	18.45	6	5	14	19	D-12	19	67	68	69	70	71	72
			19.5	19.95	6	8	6	14	D-13	14	73	74	75	76	77	78
			21.0	21.45	5	3	4	6	D-14	6	79	80	81	82	83	84
			22.5	22.95	4	2	6	8	D-15	8	85	86	87	88	89	90
			24.0	24.45	11	6	6	12	D-16	12	91	92	93	94	95	96
			25.5	25.95	7	6	12	18	D-17	18	97	98	99	100	101	102
			27.0	27.45	6	17	14	31	D-18	31	103	104	105	106	107	108
			28.5	28.95	7	14	14	28	D-19	28	109	110	111	112	113	114
			30.0	30.45	9	15	18	33	D-20	33	115	116	117	118	119	120
			31.5	31.95	13	19	20	39	D-21	39	121	122	123	124	125	126
			33.0	33.45	24	25	20	45	D-22	45	127	128	129	130	131	132
			34.5	34.95	25	25	22	47	D-23	47	133	134	135	136	137	138
			36.0	36.45	21	29	20	49	D-24	49	139	140	141	142	143	144
			37.5	37.95	27	20	30	50	D-25	50	145	146	147	148	149	150
			39.0	39.45	29	20	30	50	D-26	50	151	152	153	154	155	156
			40.5	40.95	30	25	25	50	D-27	50	157	158	159	160	161	162
									D-28	50						

RL WITH REPECT TO TOPOGRAPHY SURVEY.

0 3 6 9 12 15 18 21 24 27 30 33 36 39 42 45 48 51

0 10 20 30 40 50 60 70 80

6.0m — Light Gray 10YR (7/1), medium plasticity, soft, silty Clay, CL.

15.0m — Light Gray 10YR (7/1), medium dense, silty sand, SM.

6.0m — Gray 10YR (6/1), medium dense, Silt, SM.

15.45m — Gray 10YR (6/1), medium dense to very dense, fine Sand, SM.

LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

- PLASTIC CLAY
- SILT
- PEAT/HIGHLY ORGANIC SOIL
- UNDISTURBED SAMPLE
- MICACEOUS FINE SAND/SILT
- SAND
- SHALE
- DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^m PENETRATION= 50 BLOWS. AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED 50/X = BLOWS 50, "X" MM PENETRATION

A3	PROSOIL	SUMMERY SHEET OF TEST RESULTS	PROJECT : SRTPP-2 Geotechnical Investigation, Dhaka(Katchpur)-Sylhet-Tamabil Road(N2) 286km.

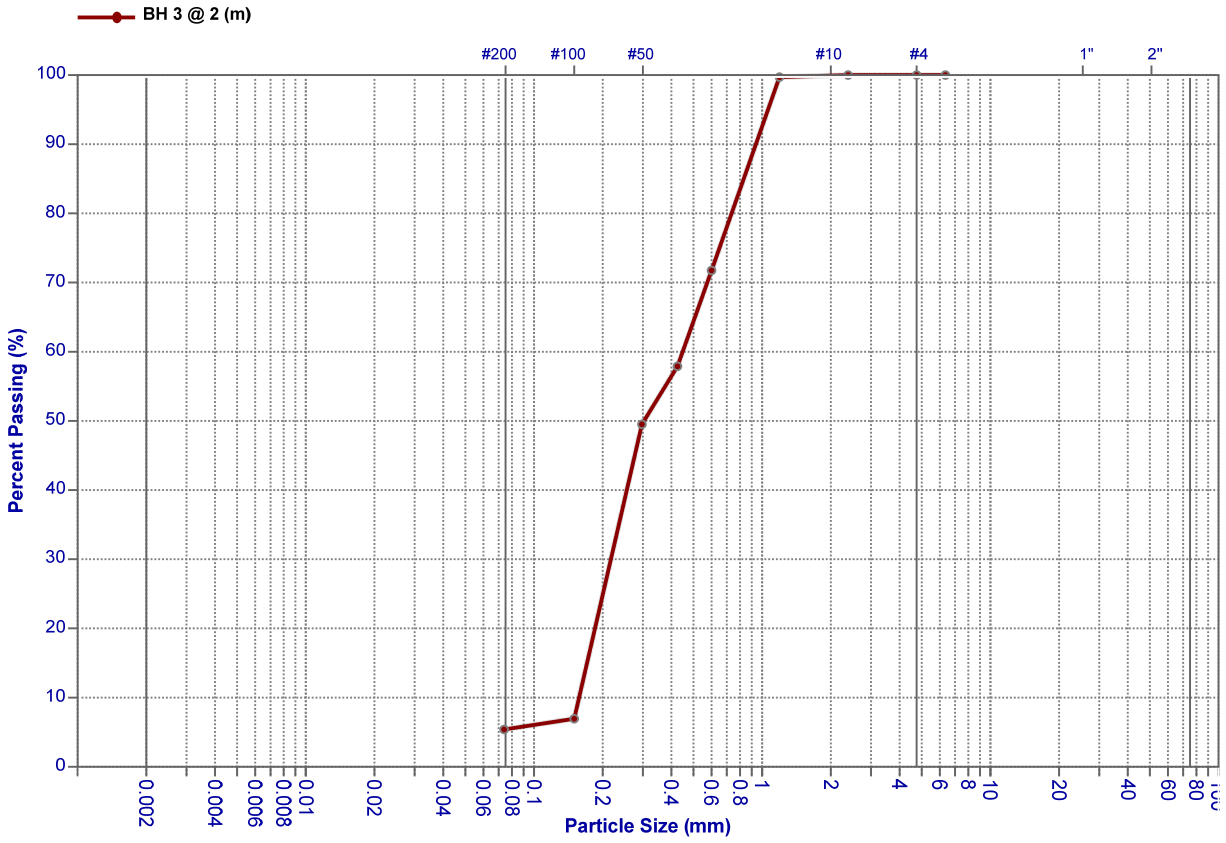
LL= Liquid limit
PL= Plastic limit
PI = Plasticity index
γ= Unit Weight (kn/m^3)
qu= Unconfined Com. Str. (kpa)
G_s= Specific gravity
c_u= Undrained shear strength (kpa)
C= Cohesion (kpa)
ϕ^o= Angle of internal friction (deg)
e_0= Void ratio
C_c= Compression index
CLH = Clay Hydrometer

BORE HOLE NO	SAMPLE	ATTERBERG LIMIT			UNIT WEIGHT	GRAIN SIZE (%)			NATURAL MOISTURE CONTENT	SPECIFIC GRAVITY	UNCONFINED COMPRESSIVE STRENGTH		CONSOLIDATION TEST		DIRECT SHEAR TEST	
		LL	PL	PI	γ (kn/m^3)	SAND	PASSING 200	CLH	% (wt)	G_s	qu	% Strain	e_0	C_c	C	ϕ^o
	UD-1															
	UD-2															
	UD-3															
	UD-4															
	D-1/1.5m	55.3	16.4	38.9					27.9	2.22						
	D-2/3.0m	56.4	14.7	41.7					29.2	2.50						
	D-3/4.5m					15.4	84.6									
	D-4/6.0m					94.6	5.4		23.2							
	D-5/7.5m									2.70						
	D-6/9.0m								5.5							
	D-7/10.5m															
	D-8/12.0m															
	D-9/13.5m								5.2							
BH 266+240	D-10/15.0m									2.46						
	D-11/16.5m															
	D-12/18.0m															
	D-13/19.5m															
	D-14/21.0m					9.5	90.5			2.31						
	D-15/22.5m					9.8	90.2									
	D-16/24.0m					7.0	93.0									
	D-17/25.5m					7.8	92.2		19.1							
	D-18/27.0m					6.9	93.1									
	D-19/28.5m					7.3	92.7									
	D-20/30.0m															
	D-21/31.5m															
	D-22/33.0m															
	D-23/34.5m															
	D-24/36.0m															
	D-25/37.5m															
	D-26/39.0m															
	D-27/40.5m															
	D-28/42.0m															
	D-29/43.5m															
	D-30/45.0m															
	D-31/46.5m															
	D-32/48.0m															
	D-33/49.5m															
	D-34/51.0m															

Sieve Analysis Test

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	6 (m)	
Job No.:	0	Classification :	SP-SM	
Location :	Sylhet (coarse sand)	Sample Type :	Disturbed	

ASTM C136



Particle Distribution

Clay	Silt	Sand	Gravel	Cobble
-	5.4	94.6	-	-

Classification

Borehole	Sample Depth (m)	D10 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	Cc	Cu	LL (%)	PI (%)	Disp. (%)	USCS	AASHTO
BH 3	2	0.158	0.217	0.303	0.448	0.665	2.835	-	-	N/A	SP-SM	A-2-4

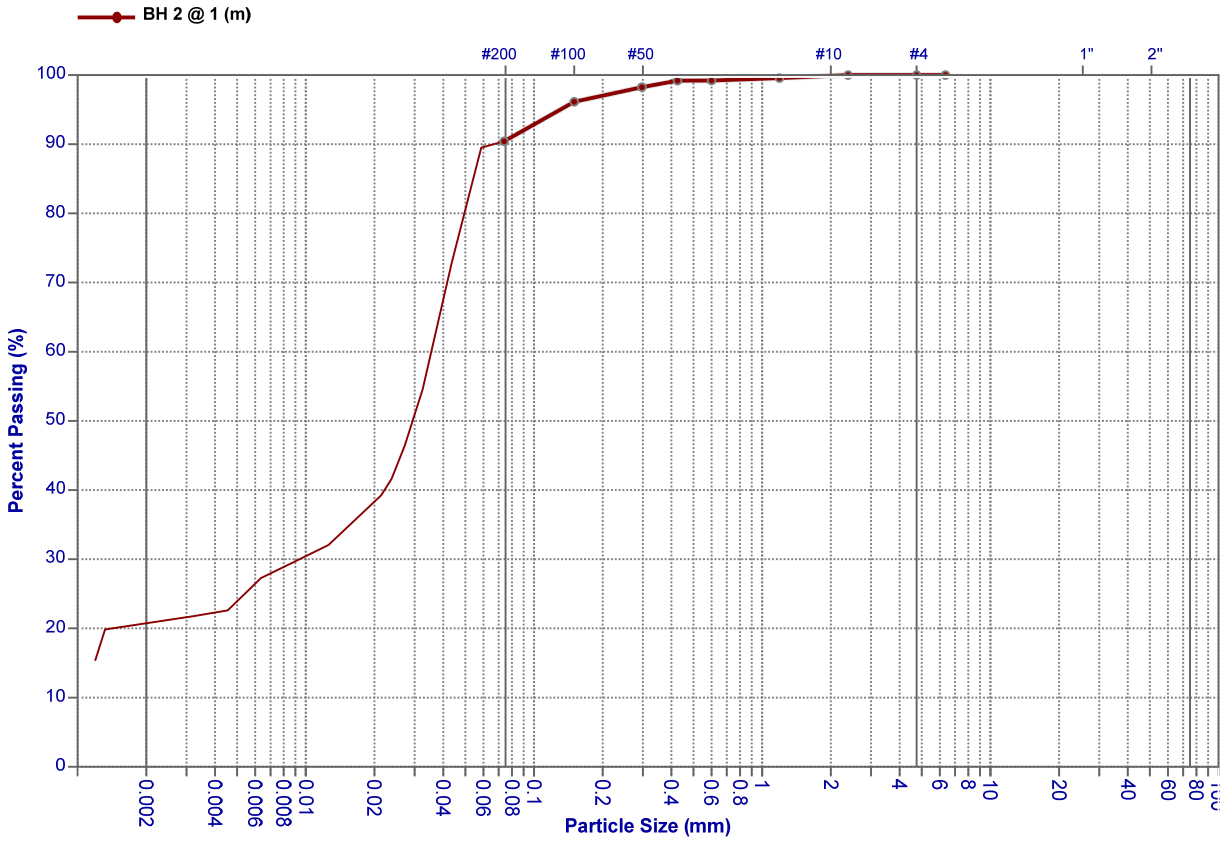
Unified Description : Poorly Graded Silty Sand With Gravel & Clay
AASHTO Description : Silty or clayey gravel and sand

Tested By :

Sieve Analysis Test

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	21 (m)	
Job No.:	0	Classification :	ML	
Location :	Norshingdi(Island 2)	Sample Type :	Disturbed	

ASTM C136



Particle Distribution

Clay	Silt	Sand	Gravel	Cobble
20.8	69.8	9.5	-	-

Classification

Borehole	Sample Depth (m)	D10 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	Cc	Cu	LL (%)	PI (%)	Disp. (%)	USCS	AASHTO
BH 2	1	-	0.009	0.029	0.035	-	-	-	-	N/A	ML	A-4

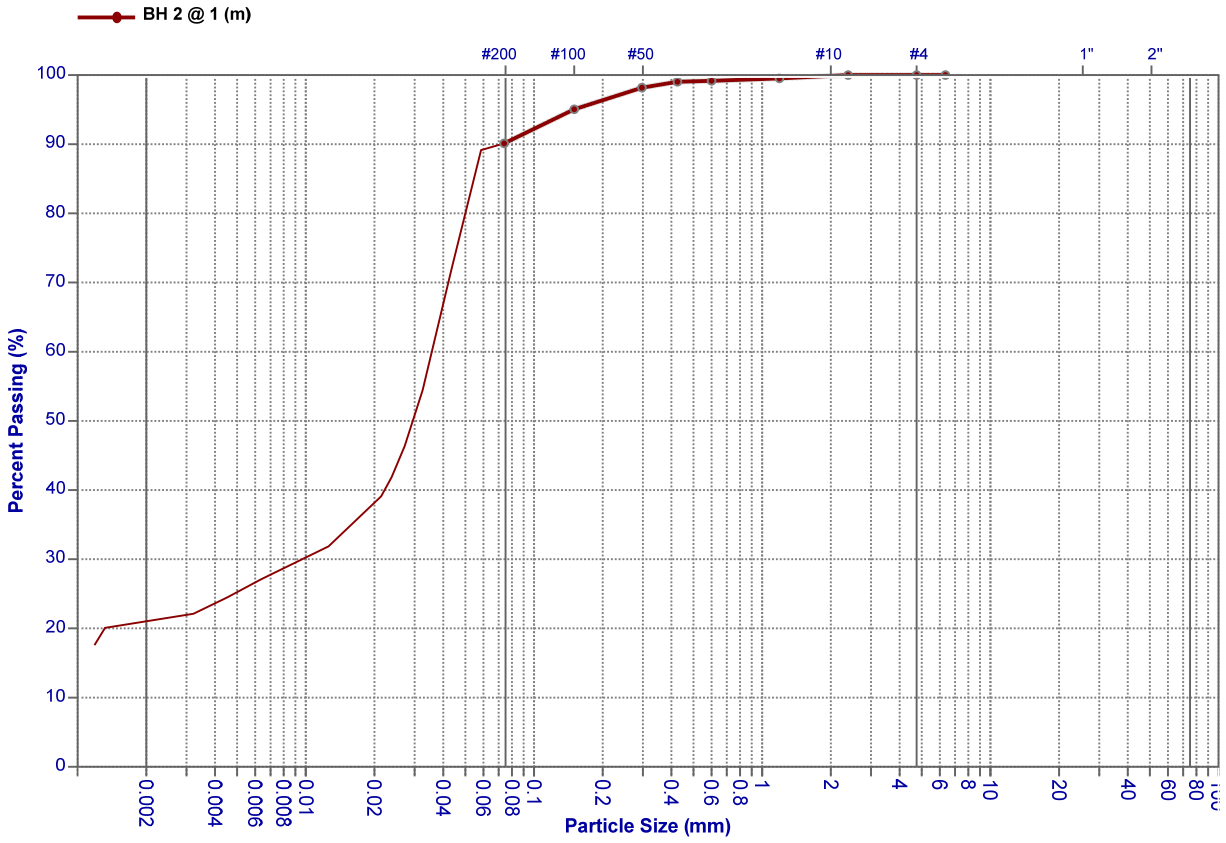
Unified Description : Low Plasticity Silt With Sand
AASHTO Description : A-4, Silty soils

Tested By :

Sieve Analysis Test

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	22.5 (m)	
Job No.:	0	Classification :	ML	
Location :	Norshingdi(Island 2)	Sample Type :	Disturbed	

ASTM C136



Particle Distribution

Clay	Silt	Sand	Gravel	Cobble
21	69.2	9.8	-	-

Classification

Borehole	Sample Depth (m)	D10 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	Cc	Cu	LL (%)	PI (%)	Disp. (%)	USCS	AASHTO
BH 2	1	-	0.01	0.029	0.036	-	-	-	-	N/A	ML	A-4

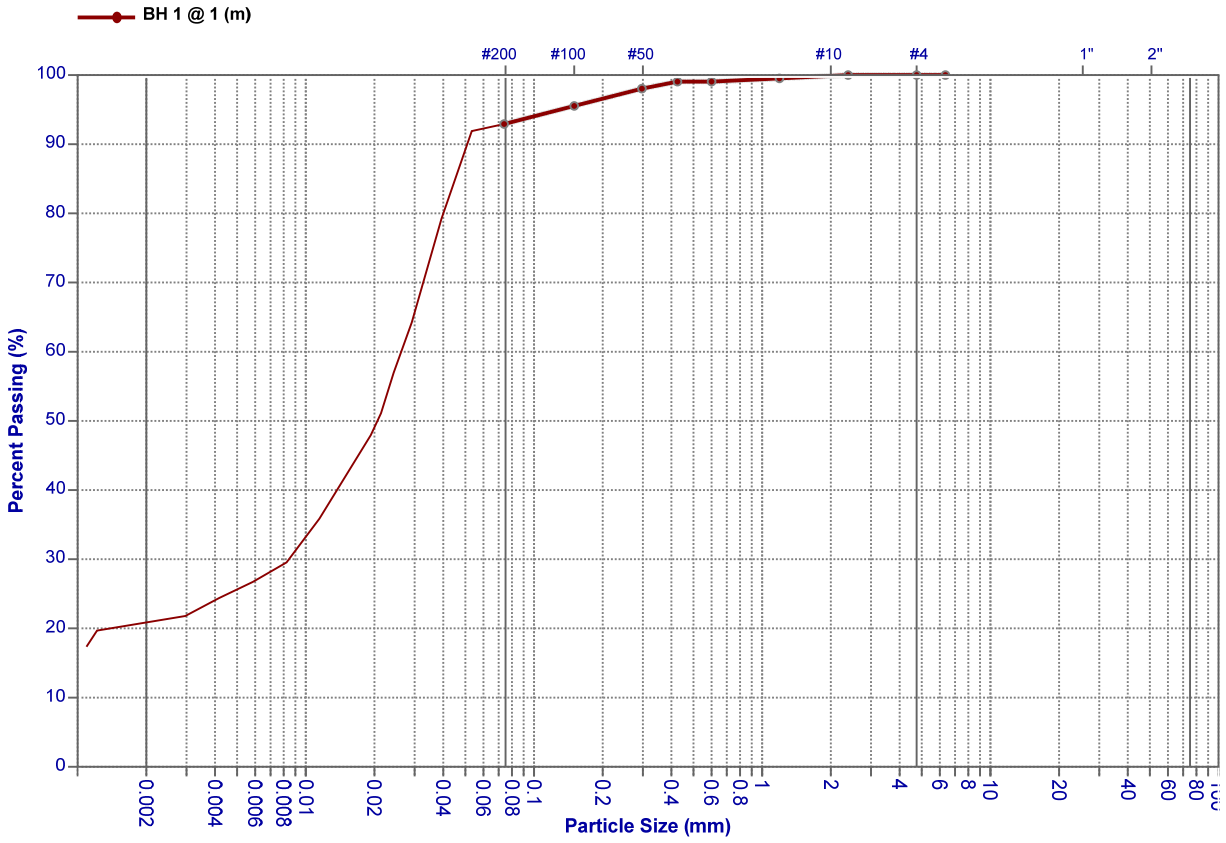
Unified Description : Low Plasticity Silt With Sand
AASHTO Description : A-4, Silty soils

Tested By :

Sieve Analysis Test

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	24 (m)	
Job No.:	0	Classification :	CL	
Location :	Norshingdi(Island)	Sample Type :	Disturbed	

ASTM C136



Particle Distribution

Clay	Silt	Sand	Gravel	Cobble
20.9	72.1	7	-	-

Classification

Borehole	Sample Depth (m)	D10 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	Cc	Cu	LL (%)	PI (%)	Disp. (%)	USCS	AASHTO
BH 1	1	-	0.008	0.021	0.026	-	-	37.9	20.3	N/A	CL	A-6

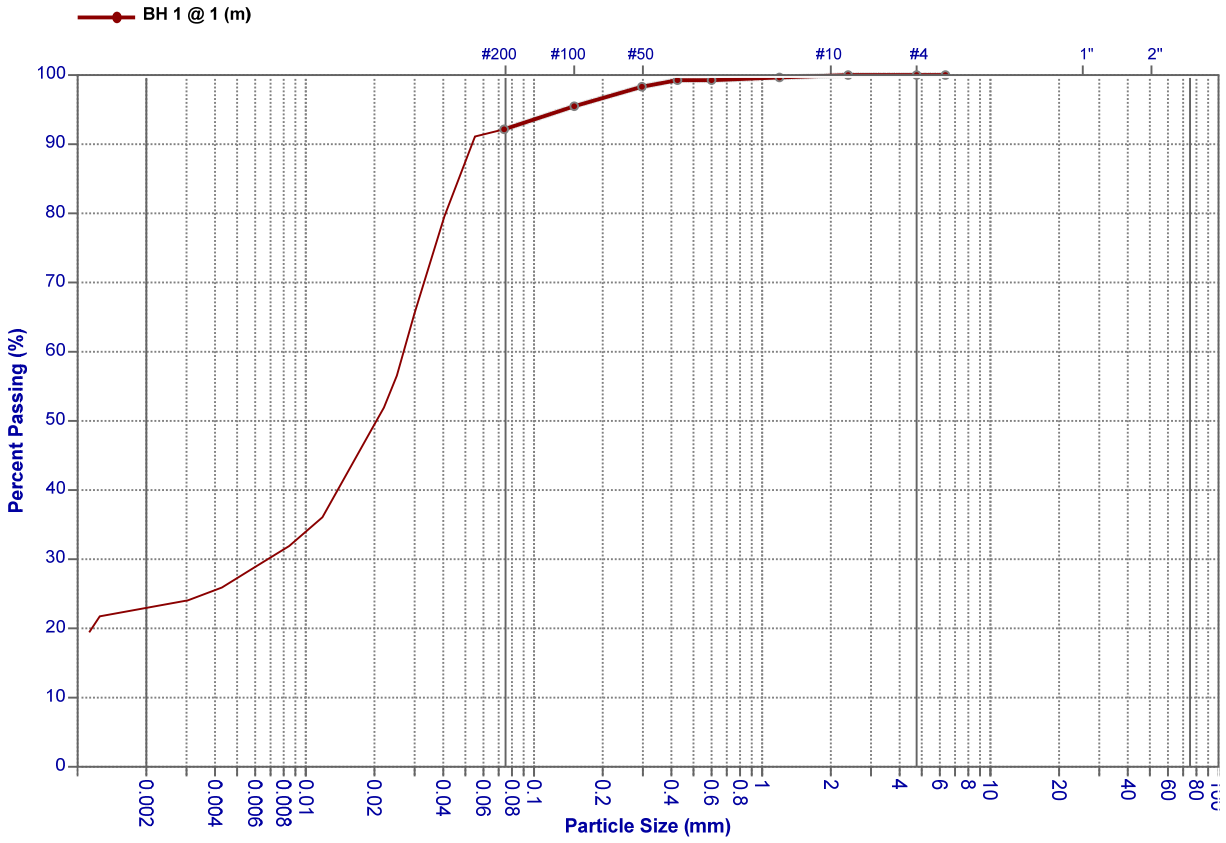
Unified Description : Low Plasticity Clay With Sand
AASHTO Description : Clayey soils

Tested By :

Sieve Analysis Test

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	25.5 (m)	
Job No.:	0	Classification :	CL	
Location :	Norshingdi(Island)	Sample Type :	Disturbed	

ASTM C136



Particle Distribution

Clay	Silt	Sand	Gravel	Cobble
23	69.2	7.8	-	-

Classification

Borehole	Sample Depth (m)	D10 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	Cc	Cu	LL (%)	PI (%)	Disp. (%)	USCS	AASHTO
BH 1	1	-	0.007	0.02	0.027	-	-	37.9	20.3	N/A	CL	A-6

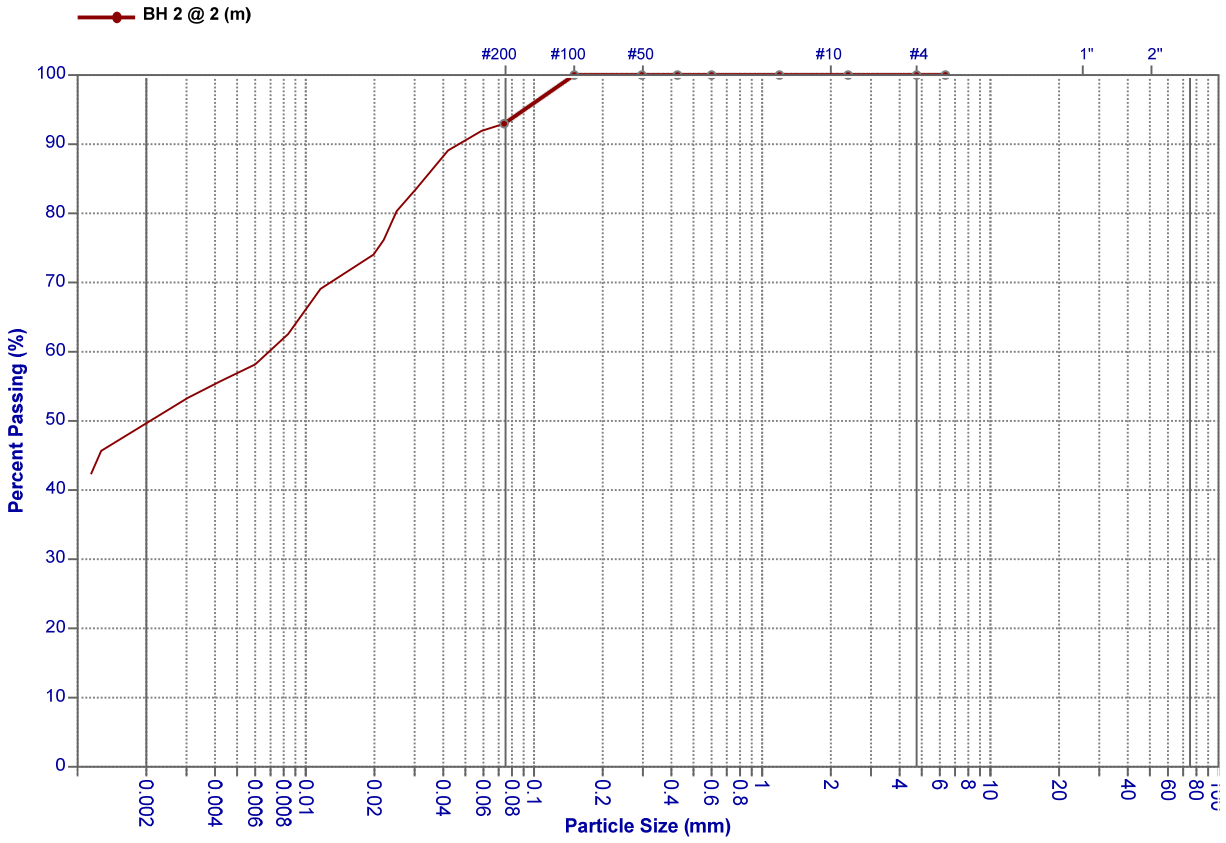
Unified Description : Low Plasticity Clay With Sand
AASHTO Description : Clayey soils

Tested By :

Sieve Analysis Test

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	1.5 (m)	
Job No.:	0	Classification :	CH	
Location :	Gutiabo(red clay)	Sample Type :	Disturbed	

ASTM C136



Particle Distribution

Clay	Silt	Sand	Gravel	Cobble
49.7	43.4	6.9	-	-

Classification

Borehole	Sample Depth (m)	D10 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	Cc	Cu	LL (%)	PI (%)	Disp. (%)	USCS	AASHTO
BH 2	2	-	-	0.002	0.007	-	-	55.3	38.9	N/A	CH	A-7-6

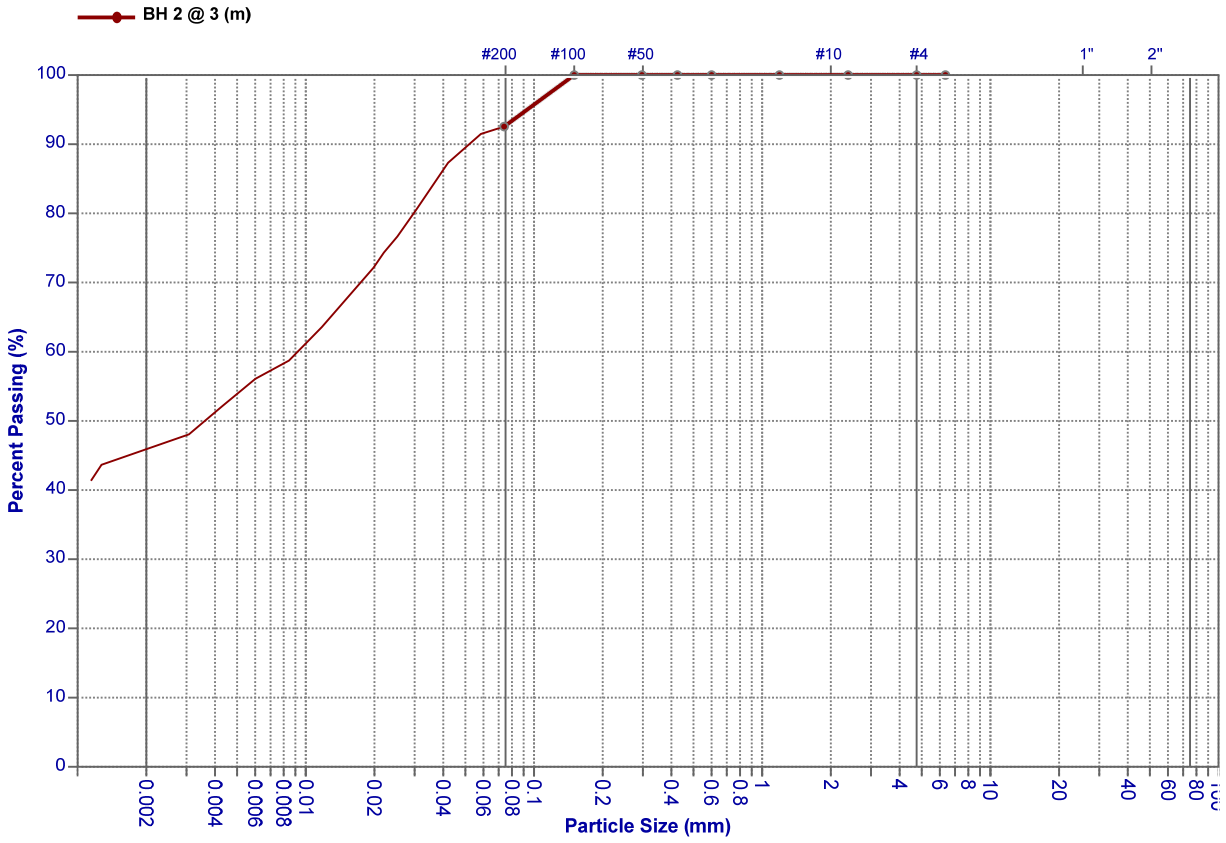
Unified Description : High Plasticity Clay With Sand
AASHTO Description : Clayey soils

Tested By :

Sieve Analysis Test

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	3 (m)	
Job No.:	0	Classification :	CH	
Location :	Gutiabo(red clay)	Sample Type :	Disturbed	

ASTM C136



Particle Distribution

Clay	Silt	Sand	Gravel	Cobble
45.9	46.8	7.3	-	-

Classification

Borehole	Sample Depth (m)	D10 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	Cc	Cu	LL (%)	PI (%)	Disp. (%)	USCS	AASHTO
BH 2	3	-	-	0.004	0.009	-	-	56.4	41.7	N/A	CH	A-7-6

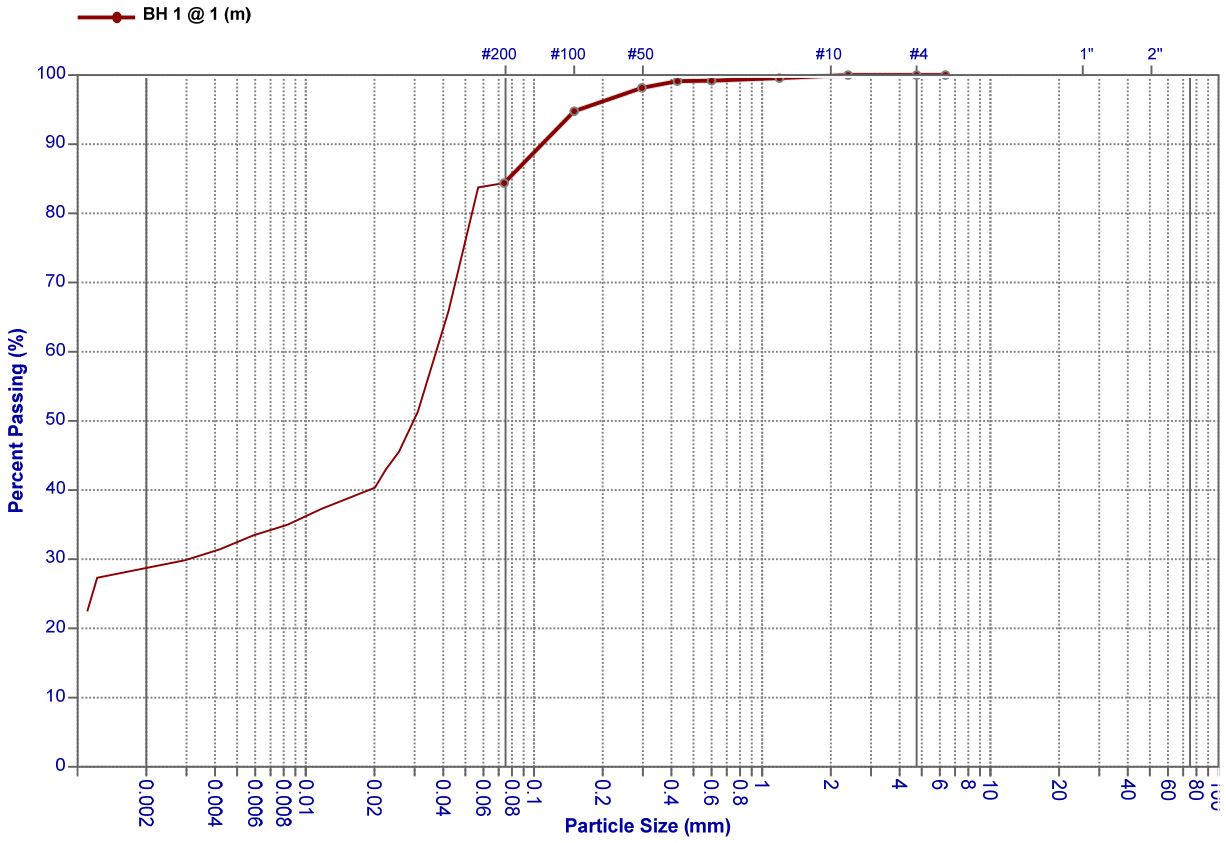
Unified Description : High Plasticity Clay With Sand
AASHTO Description : Clayey soils

Tested By :

Sieve Analysis Test

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	4.5 (m)	
Job No.:	0	Classification :	ML	
Location :	Norshingdi(west bank)	Sample Type :	Disturbed	

ASTM C136



Particle Distribution

Clay	Silt	Sand	Gravel	Cobble
28.7	55.8	15.4	-	-

Classification

Borehole	Sample Depth (m)	D10 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	Cc	Cu	LL (%)	PI (%)	Disp. (%)	USCS	AASHTO
BH 1	1	-	0.003	0.03	0.037	-	-	-	-	N/A	ML	A-4

Unified Description : Low Plasticity Silt With Sand
AASHTO Description : A-4, Silty soils

Tested By :

ATTERBERG LIMITS TEST

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	1.5 (m)	
Job No.:	0	Classification :	CH	
Location :	Gutiabo(red clay)	Sample Type :	Disturbed	

ASTM D2216-90, D854

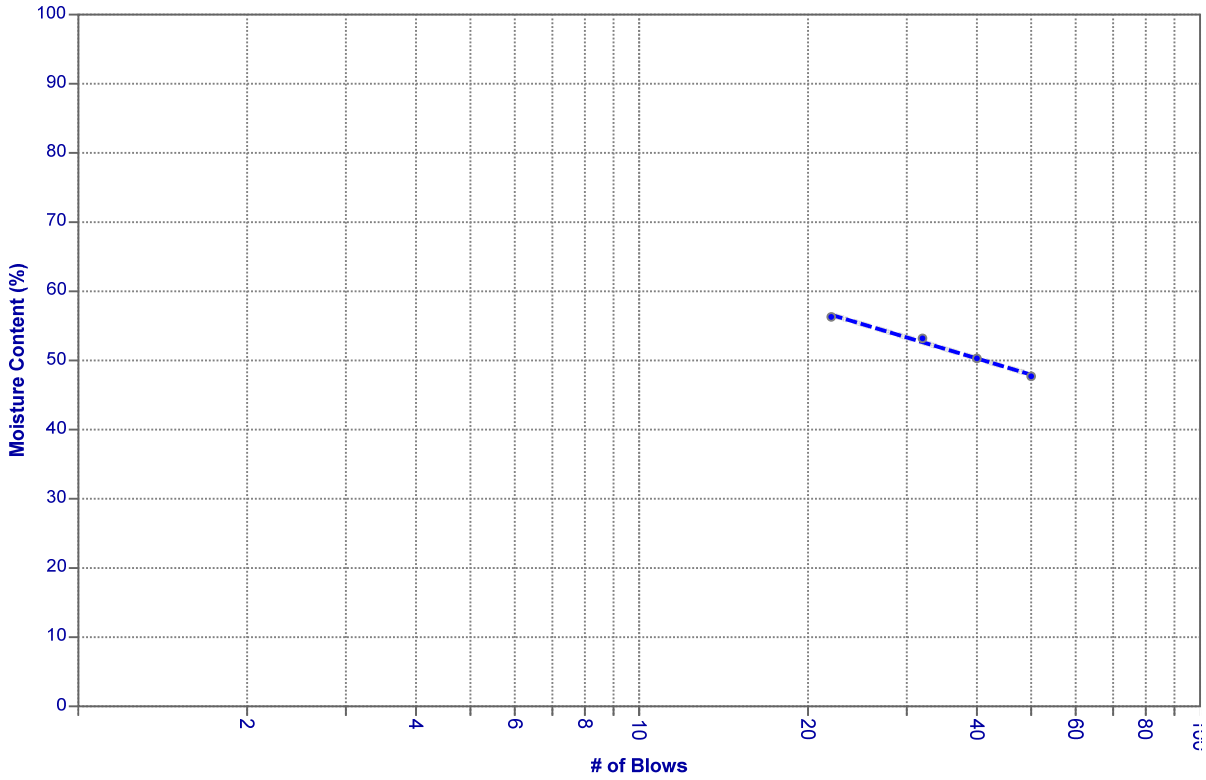
Liquid Limit

Cont. W (gr)	Cont. + Wet Soil W (gr)	Cont. + Dry Soil W (gr)	# of Blows	Moisture Content (%)
38.07	50.95	46.31	22	56.3
30.57	40.02	36.74	32	53.2
30.05	39.46	36.31	40	50.3
29.53	42.23	38.13	50	47.7
				-

Plastic Limit

Cont. W (gr)	Cont. + Wet Soil W (gr)	Cont. + Dry Soil W (gr)	Moisture Content (%)
37.86	59.42	56.12	18.1
37	60.63	57.6	14.7

LL = 55.3 % **PL = 16.4 %** **PI=38.9**



Unified Description : High Plasticity Clay With Sand
AASHTO Description : Clayey soils

Tested By :

ATTERBERG LIMITS TEST

Project :	Subregional Transport Project	Borehole :	BH 266+240	Prosoil
Client :	Hifab International	Sample Depth :	3 (m)	
Job No.:	0	Classification :	CH	
Location :	Gutiabo(red clay)	Sample Type :	Disturbed	

ASTM D2216-90, D854

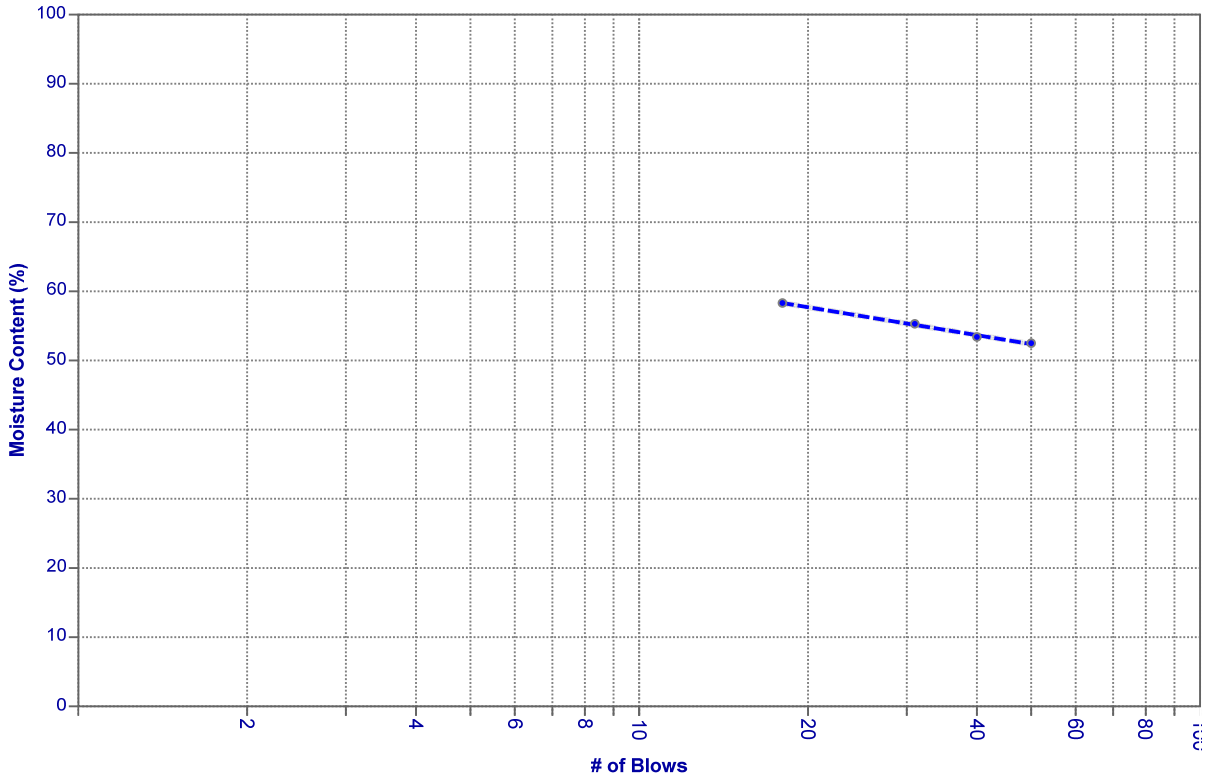
Liquid Limit

Cont. W (gr)	Cont. + Wet Soil W (gr)	Cont. + Dry Soil W (gr)	# of Blows	Moisture Content (%)
39.55	50.85	46.69	18	58.3
36.4	45.25	42.1	31	55.3
30.9	39.4	36.44	40	53.4
37	46.53	43.25	50	52.5
				-

Plastic Limit

Cont. W (gr)	Cont. + Wet Soil W (gr)	Cont. + Dry Soil W (gr)	Moisture Content (%)
38.94	52.81	51.04	14.6
30.57	42.99	41.4	14.7

LL = 56.4 % **PL = 14.7 %** **PI=41.7**



Unified Description : High Plasticity Clay With Sand
AASHTO Description : Clayey soils

Tested By :

DATA SHEET

MOISTURE CONTENT

Project : Subregional Transport Project

B.H. No.: BH 266+240 Sample no.: 6 Depth: 9.0m Test date : 20/9/2014

Determination No.	1	2	3
Can No.	22	33	34
Wt. of Can + wet soil (g)	52.77	69.49	219.29
Wt. of Can + dry soil (g)	51.21	68.72	210.19
Wt. of Can (g)	30.93	44.53	46.67
Wt. of Water (g)	1.56	0.77	9.1
Wt. of dry soil (g)	20.28	24.19	163.52
Water Content, w (%)	7.69	3.18	5.57

Signature of Technician

Signature of Engineer

DATA SHEET

MOISTURE CONTENT

Project : Subregional Transport Project

B.H. No.: BH 266+240 Sample no.: 9 Depth: 13.5m Test date : 20/9/2014

Determination No.	1	2	3
Can No.	9	21	1
Wt. of Can + wet soil (g)	84.45	81.94	74.84
Wt. of Can + dry soil (g)	83	80.26	73.64
Wt. of Can (g)	54.56	49.3	50.22
Wt. of Water (g)	1.45	1.68	1.2
Wt. of dry soil (g)	28.44	30.96	23.42
Water Content, w (%)	5.10	5.43	5.12

Signature of Technician

Signature of Engineer

DATA SHEET

MOISTURE CONTENT

Project : Subregional Transport Project

B.H. No.: BH 266+240 Sample no.: 01 Depth: 1.5m Test date : 20/9/2014

Determination No.	1	2	3
Can No.	15	28	51
Wt. of Can + wet soil (g)	78.67	93.97	91.53
Wt. of Can + dry soil (g)	71.24	85.61	83.16
Wt. of Can (g)	44.65	55.58	53.17
Wt. of Water (g)	7.43	8.36	8.37
Wt. of dry soil (g)	26.59	30.03	29.99
Water Content, w (%)	27.94	27.84	27.91

Signature of Technician

Signature of Engineer

DATA SHEET

MOISTURE CONTENT

Project : Subregional Transport Project

B.H. No.: BH 266+240 Sample no.: 04 Depth: 6.0 Test date : 20/9/2014

Determination No.	1	2	3
Can No.	6	26	14
Wt. of Can + wet soil (g)	72.95	72.09	72.92
Wt. of Can + dry soil (g)	66.63	65.95	64.54
Wt. of Can (g)	39.36	37.93	30.44
Wt. of Water (g)	6.32	6.14	8.38
Wt. of dry soil (g)	27.27	28.02	34.1
Water Content, w (%)	23.18	21.91	24.57

Signature of Technician

Signature of Engineer

DATA SHEET

MOISTURE CONTENT

Project : Subregional Transport Project

B.H. No.: BH 266+240 Sample no.: 02 Depth: 3.0m Test date : 20/9/2014

Determination No.	1	2	3
Can No.	20	22	47
Wt. of Can + wet soil (g)	62.15	65.22	82.89
Wt. of Can + dry soil (g)	54.99	57.1	75.03
Wt. of Can (g)	30.47	31	48.09
Wt. of Water (g)	7.16	8.12	7.86
Wt. of dry soil (g)	24.52	27.7	26.94
Water Content, w (%)	29.20	29.31	29.18

Signature of Technician

Signature of Engineer

DATA SHEET

MOISTURE CONTENT

Project : Subregional Transport Project

B.H. No.: BH 266+240 Sample no.: 17 Depth: 25.5m Test date : 20/9/2014

Determination No.	1	2	3
Can No.	12	28	32
Wt. of Can + wet soil (g)	50.42	75.47	66.58
Wt. of Can + dry soil (g)	47.46	72.33	63.85
Wt. of Can (g)	31.49	55.58	50.22
Wt. of Water (g)	2.96	3.14	2.73
Wt. of dry soil (g)	15.97	16.75	13.63
Water Content, w (%)	18.53	18.75	20.03

Signature of Technician

Signature of Engineer

SPECIFIC GRAVITY OF SOIL

Project: Subregional Transport Project

BH: BH 266+240 Type: sand Depth:7.5m

Number of volumetric flash	10	11	
Wt. of volumetric flash (g)			
Wt. of volumetric flash + Water (g)	342.52	340.51	
Water Temperature in vol.flash before test (t0C)	31.4	31.4	
Specific gravity of water at t0C (g/cm3)	0.9953	0.9953	
Wt. of dry soil in volumetric flash (g)	41.35	40.28	
Wt. of volum.flash + Water+ drysoil (g)	369.02	365.52	
Water Temperature in vol.flash after test (t0C)	31.4	31.4	
Specific gravity of water at t0C	0.9953	0.9953	
Wt. volumetric flash + Water at Temp t0C	342.52	340.51	
Specific gravity of soil, Gs (g/cm3)	2.771	2.625	
Average specific gravity of soil, Gs (g/cm3)		2.698	

Specific Gravity of Water

water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3
4	1	16	0.9989	28	0.9962
5	1	17	0.9988	29	0.9959
6	0.9999	18	0.9986	30	0.9957
7	0.9999	19	0.9984	31	0.9953
8	0.9999	20	0.9982	32	0.995
9	0.9998	21	0.998	33	0.9947
10	0.9997	22	0.9978	34	0.9944
11	0.9996	23	0.9975	35	0.994
12	0.9995	24	0.9973	36	0.9937
13	0.9994	25	0.997	37	0.9933
14	0.9992	26	0.9968	38	0.993
15	0.9991	27	0.9965	39	0.9926

SPECIFIC GRAVITY OF SOIL

Project: Subregional Transport Project

BH: BH 266+240 Type:clay Depth:3.0m

Number of volumetric flash	10	11	12
Wt. of volumetric flash (g)			
Wt. of volumetric flash + Water (g)	337.16	341.24	346.14
Water Temperature in vol.flash before test (t0C)	31	31	31
Specific gravity of water at t0C (g/cm3)	0.9953	0.9953	0.9953
Wt. of dry soil in volumetric flash (g)	31.96	31.06	30.47
Wt. of volum.flash + Water+ drysoil (g)	356.68	359.81	364.38
Water Temperature in vol.flash after test (t0C)	31	31	31
Specific gravity of water at t0C	0.9953	0.9953	0.9953
Wt. volumetric flash + Water at Temp t0C	337.16	341.24	346.14
Specific gravity of soil, Gs (g/cm3)	2.557	2.475	2.480
Average specific gravity of soil, Gs (g/cm3)		2.504	

Specific Gravity of Water

water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3
4	1	16	0.9989	28	0.9962
5	1	17	0.9988	29	0.9959
6	0.9999	18	0.9986	30	0.9957
7	0.9999	19	0.9984	31	0.9953
8	0.9999	20	0.9982	32	0.995
9	0.9998	21	0.998	33	0.9947
10	0.9997	22	0.9978	34	0.9944
11	0.9996	23	0.9975	35	0.994
12	0.9995	24	0.9973	36	0.9937
13	0.9994	25	0.997	37	0.9933
14	0.9992	26	0.9968	38	0.993
15	0.9991	27	0.9965	39	0.9926

SPECIFIC GRAVITY OF SOIL

Project: Subregional Transport Project

BH: BH 266+240 Type:silty sand Depth:21m

Number of volumetric flash	10	11	12
Wt. of volumetric flash (g)			
Wt. of volumetric flash + Water (g)	351.19	324.75	361.98
Water Temperature in vol.flash before test (t0C)	31	31	31
Specific gravity of water at t0C (g/cm3)	0.9953	0.9953	0.9953
Wt. of dry soil in volumetric flash (g)	30.77	30.43	30.43
Wt. of volum.flash + Water+ drysoil (g)	368.81	341.94	379.33
Water Temperature in vol.flash after test (t0C)	31	31	31
Specific gravity of water at t0C	0.9953	0.9953	0.9953
Wt. volumetric flash + Water at Temp t0C	351.19	324.75	361.98
Specific gravity of soil, Gs (g/cm3)	2.329	2.288	2.316
Average specific gravity of soil, Gs (g/cm3)		2.311	

Specific Gravity of Water

water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3
4	1	16	0.9989	28	0.9962
5	1	17	0.9988	29	0.9959
6	0.9999	18	0.9986	30	0.9957
7	0.9999	19	0.9984	31	0.9953
8	0.9999	20	0.9982	32	0.995
9	0.9998	21	0.998	33	0.9947
10	0.9997	22	0.9978	34	0.9944
11	0.9996	23	0.9975	35	0.994
12	0.9995	24	0.9973	36	0.9937
13	0.9994	25	0.997	37	0.9933
14	0.9992	26	0.9968	38	0.993
15	0.9991	27	0.9965	39	0.9926

SPECIFIC GRAVITY OF SOIL

Project: Subregional Transport Project

BH: BH 266+240 Type:clay Depth:1.5m

Number of volumetric flash	10	11	12
Wt. of volumetric flash (g)			
Wt. of volumetric flash + Water (g)	326.7	335.64	349.38
Water Temperature in vol.flash before test (t0C)	28	28	28
Specific gravity of water at t0C (g/cm3)	0.9962	0.9962	0.9962
Wt. of dry soil in volumetric flash (g)	30.94	30.37	31.21
Wt. of volum.flash + Water+ drysoil (g)	344.21	353.14	365.04
Water Temperature in vol.flash after test (t0C)	28	28	28
Specific gravity of water at t0C	0.9962	0.9962	0.9962
Wt. volumetric flash + Water at Temp t0C	326.7	335.64	349.38
Specific gravity of soil, Gs (g/cm3)	2.295	2.351	1.999
Average specific gravity of soil, Gs (g/cm3)		2.215	

Specific Gravity of Water

water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3
4	1	16	0.9989	28	0.9962
5	1	17	0.9988	29	0.9959
6	0.9999	18	0.9986	30	0.9957
7	0.9999	19	0.9984	31	0.9953
8	0.9999	20	0.9982	32	0.995
9	0.9998	21	0.998	33	0.9947
10	0.9997	22	0.9978	34	0.9944
11	0.9996	23	0.9975	35	0.994
12	0.9995	24	0.9973	36	0.9937
13	0.9994	25	0.997	37	0.9933
14	0.9992	26	0.9968	38	0.993
15	0.9991	27	0.9965	39	0.9926

SPECIFIC GRAVITY OF SOIL

Project: Subregional Transport Project

BH: BH 266+240 Type:silty sand Depth:15m

Number of volumetric flash	10	11	12
Wt. of volumetric flash (g)			
Wt. of volumetric flash + Water (g)	326.7	335.64	349.38
Water Temperature in vol.flash before test (t0C)	30.8	30.8	30.8
Specific gravity of water at t0C (g/cm3)	0.9957	0.9957	0.9957
Wt. of dry soil in volumetric flash (g)	31.3	30.42	30.43
Wt. of volum.flash + Water+ drysoil (g)	345.1	353.79	367.7
Water Temperature in vol.flash after test (t0C)	30.8	30.8	30.8
Specific gravity of water at t0C	0.9957	0.9957	0.9957
Wt. volumetric flash + Water at Temp t0C	326.7	335.64	349.38
Specific gravity of soil, Gs (g/cm3)	2.416	2.469	2.502
Average specific gravity of soil, Gs (g/cm3)		2.462	

Specific Gravity of Water

water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3	water T (^{0}C)	S.Gravity g/cm^3
4	1	16	0.9989	28	0.9962
5	1	17	0.9988	29	0.9959
6	0.9999	18	0.9986	30	0.9957
7	0.9999	19	0.9984	31	0.9953
8	0.9999	20	0.9982	32	0.995
9	0.9998	21	0.998	33	0.9947
10	0.9997	22	0.9978	34	0.9944
11	0.9996	23	0.9975	35	0.994
12	0.9995	24	0.9973	36	0.9937
13	0.9994	25	0.997	37	0.9933
14	0.9992	26	0.9968	38	0.993
15	0.9991	27	0.9965	39	0.9926

Bored Pile Ultimate Capacity Calculation

Pile diameter 1200 mm

Project Road Name:					SRTPPF-2 Geotechnical Investigation, Dhaka(Katchpur)-Sylhet-Tamabil Road(N2) 283km.														Pile Diameter (m):		1.200
Bridge ID:					266+240				Borehole ID:		BH NO: 266+240								Surface Area per lm(sqm):		3.771
Bridge Loacation:					Name of place: Dhaka-Sylhet-Tamabil Road.														Pile Toe Area (sqm) :		1.131
SPT Energy correction Factor : (Trip Hammer used)				1.00			qs=Ks*Po*tan(δ)						Dc=	24		Qb=		P0*Nq*Ab=			13,577
Depth, m	Red. Level	Soil type	N-value		γ	ϕ	c	α	K_0=1-sinϕ	K_s/k_0	ks	tan(δ)	γ_{sub}	p_0= γD,	Shaft Resistance ultimate						Total ultimate
			Field	Corr'd SPT											Unit q_s	Corr q_s	qs(segment)	Q_s	Q_b		Q T
	PWD			N60	kN/cum		kPa						kN/cum	kN/sqm	kPa	kPa	kN	kN	kN		kN
0	-14.70																				
1.50	(16.20)	c	4	4	14.75	FALSE	20	0.70	1.00	0.85	0.85	FALSE	4.75	7	14	14	79	**79**			
3.00	(17.70)	c	3	3	14.50	FALSE	15	0.70	1.00	0.85	0.85	FALSE	4.50	14	11	11	59	**139**			
4.50	(19.20)	c	3	3	14.50	FALSE	15	0.70	1.00	0.85	0.85	FALSE	4.50	20	11	11	59	**198**			
6.00	(20.70)	s	9	9	16.50	27.70	-	0.70	0.535375	0.85	0.46	0.45	6.50	39	8	8	45	**243**			
7.50	(22.20)	s	15	15	17.75	29.50	-	0.70	0.51	0.85	0.43	0.45	7.75	58	11	11	64	**307**			
9.00	(23.70)	s	22	22	18.53	31.60	-	0.70	0.48	0.85	0.40	0.45	8.53	77	14	14	79	**386**			
10.50	(25.20)	s	18	18	18.25	30.40	-	0.70	0.49	0.85	0.42	0.45	8.25	87	16	16	93	**479**			
12.00	(26.70)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	101	19	19	107	**586**			
13.50	(28.20)	s	17	17	18.15	30.10	-	0.70	0.50	0.85	0.42	0.45	8.15	110	21	21	119	**705**			
15.00	(29.70)	s	14	14	17.52	29.20	-	0.70	0.51	0.85	0.44	0.45	7.52	113	22	22	125	**830**			
16.50	(31.20)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	139	26	26	148	**978**			
18.00	(32.70)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	152	28	28	161	**1,139**			
19.50	(34.20)	s	14	14	17.52	29.20	-	0.70	0.51	0.85	0.44	0.45	7.52	147	29	29	163	**1,302**			
21.00	(35.70)	s	6	6	15.75	26.80	-	0.70	0.55	0.85	0.47	0.45	5.75	121	25	25	144	**1,445**			
22.50	(37.20)	s	8	8	16.25	27.40	-	0.70	0.54	0.85	0.46	0.45	6.25	141	29	29	164	**1,609**			
24.00	(38.70)	s	12	12	17.30	28.60	-	0.70	0.52	0.85	0.44	0.45	7.30	175	35	35	198	**1,807**			
25.50	(40.20)	s	18	18	18.25	30.40	-	0.70	0.49	0.85	0.42	0.45	8.25	175	33	33	187	**1,994**			
27.00	(41.70)	s	31	31	19.10	34.30	-	0.70	0.44	0.85	0.37	0.45	9.10	175	29	29	166	**2,160**			
28.50	(43.20)	s	28	28	18.95	33.40	-	0.70	0.45	0.85	0.38	0.45	8.95	175	30	30	171	**2,331**			
30.00	(44.70)	s	33	33	19.25	34.90	-	0.70	0.43	0.85	0.36	0.70	9.25	175	44	44	251	**2,582**			
31.50	(46.20)	s	39	39	19.49	36.70	-	0.70	0.40	0.85	0.34	0.74	9.49	175	45	45	253	**2,835**			
33.00	(47.70)	s	45	45	20.25	38.50	-	0.70	0.38	0.85	0.32	0.79	10.25	175	45	45	253	**3,088**			
34.50	(49.20)	s	47	47	21.00	39.10	-	1.70	0.37	1.85	0.68	0.81	11.00	175	97	97	550	**3,638**			
36.00	(50.70)	s	49	49	21.75	39.70	-	2.70	0.36	2.85	1.03	0.83	11.75	175	150	110	622	**4,260**			
37.50	(52.20)	s	50	50	22.00	40.00	-	3.70	0.36	3.85	1.38	0.84	12.00	175	202	110	622	**4,883**			
39.00	(53.70)	s	50	50	22.00	40.00	-	4.70	0.36	4.85	1.73	0.84	12.00	175	255	110	622	**5,505**			
40.50	(55.20)	s	50	50	22.00	40.00	-	5.70	0.36	5.85	2.09	0.84	12.00	175	307	110	622	**6,127**			
42.00	(56.70)	s	50	50	22.00	40.00	-	6.70	0.36	6.85	2.45	0.84	12.00	175	360	110	622	**6,749**			

Bored Pile Ultimate Capacity Calculation

Pile diameter 1000 mm

Project Road Name:	SRTPPF-2 Geotechnical Investigation, Dhaka(Katchpur)-Sylhet-Tamabil Road(N2) 283km.			Pile Diameter (m):	1.000
Bridge ID:	266+240	Borehole ID:	BH NO: 266+240	Surface Area per lm(sqm):	3.143
Bridge Loacation:	Name of place: Dhaka-Sylhet-Tamabil Road.			Pile Toe Area (sqm) :	0.786
SPT Energy correction Factor : (Trip Hammer used) 1.00	qs=Ks*Po*tan(δ)		Dc= 20	Qb= P0*Nq*Ab=	7,857

Depth, m	Red. Level	Soil type	N-value		γ	ϕ	c	α	K_0=1-sinϕ	K_s/k_0	ks	tan(δ)	γ_{sub}	$p_0=\gamma D$,	Shaft Resistance ultimate				Q_b	Total ultimate
			Field	Corr'd SPT											Unit q_s	Corr q_s	qs(segment)	Q_s		Q T
	PWD			N60	kN/cum		kPa						kN/cum	kN/sqm	kPa	kPa	kN	kN	kN	kN
0	-14.70																			
1.50	(16.20)	c	4	4	14.75	FALSE	20	0.70	1.00	0.85	0.85	FALSE	4.75	7	14	14	66	**66**		
3.00	(17.70)	c	3	3	14.50	FALSE	15	0.70	1.00	0.85	0.85	FALSE	4.50	14	11	11	50	**116**		
4.50	(19.20)	c	3	3	14.50	FALSE	15	0.70	1.00	0.85	0.85	FALSE	4.50	20	11	11	50	**165**		
6.00	(20.70)	s	9	9	16.50	27.70	-	0.70	0.54	0.85	0.46	0.45	6.50	39	8	8	38	**203**		
7.50	(22.20)	s	15	15	17.75	29.50	-	0.70	0.51	0.85	0.43	0.45	7.75	58	11	11	53	**256**		
9.00	(23.70)	s	22	22	18.53	31.60	-	0.70	0.48	0.85	0.40	0.45	8.53	77	14	14	66	**322**		
10.50	(25.20)	s	18	18	18.25	30.40	-	0.70	0.49	0.85	0.42	0.45	8.25	87	16	16	77	**399**		
12.00	(26.70)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	101	19	19	90	**489**		
13.50	(28.20)	s	17	17	18.15	30.10	-	0.70	0.50	0.85	0.42	0.45	8.15	110	21	21	99	**587**		
15.00	(29.70)	s	14	14	17.52	29.20	-	0.70	0.51	0.85	0.44	0.45	7.52	113	22	22	104	**692**		
16.50	(31.20)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	139	26	26	123	**815**		
18.00	(32.70)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	152	28	28	134	**949**		
19.50	(34.20)	s	14	14	17.52	29.20	-	0.70	0.51	0.85	0.44	0.45	7.52	147	29	29	135	**1,085**		
21.00	(35.70)	s	6	6	15.75	26.80	-	0.70	0.55	0.85	0.47	0.45	5.75	147	31	31	145	**1,230**		
22.50	(37.20)	s	8	8	16.25	27.40	-	0.70	0.54	0.85	0.46	0.45	6.25	147	30	30	143	**1,373**		
24.00	(38.70)	s	12	12	17.30	28.60	-	0.70	0.52	0.85	0.44	0.45	7.30	147	29	29	138	**1,511**		
25.50	(40.20)	s	18	18	18.25	30.40	-	0.70	0.49	0.85	0.42	0.45	8.25	147	28	28	131	**1,641**		
27.00	(41.70)	s	31	31	19.10	34.30	-	0.70	0.44	0.85	0.37	0.45	9.10	147	24	24	115	**1,757**		
28.50	(43.20)	s	28	28	18.95	33.40	-	0.70	0.45	0.85	0.38	0.45	8.95	147	25	25	119	**1,876**		
30.00	(44.70)	s	33	33	19.25	34.90	-	0.70	0.43	0.85	0.36	0.70	9.25	147	37	37	175	**2,051**		
31.50	(46.20)	s	39	39	19.49	36.70	-	0.70	0.40	0.85	0.34	0.74	9.49	147	37	37	176	**2,227**		
33.00	(47.70)	s	45	45	20.25	38.50	-	0.70	0.38	0.85	0.32	0.79	10.25	147	37	37	176	**2,404**		
34.50	(49.20)	s	47	47	21.00	39.10	-	1.70	0.37	1.85	0.68	0.81	11.00	147	81	81	384	**2,788**		
36.00	(50.70)	s	49	49	21.75	39.70	-	2.70	0.36	2.85	1.03	0.83	11.75	147	125	110	519	**3,306**		
37.50	(52.20)	s	50	50	22.00	40.00	-	3.70	0.36	3.85	1.38	0.84	12.00	147	169	110	519	**3,825**		
39.00	(53.70)	s	50	50	22.00	40.00	-	4.70	0.36	4.85	1.73	0.84	12.00	147	213	110	519	**4,343**		
40.50	(55.20)	s	50	50	22.00	40.00	-	5.70	0.36	5.85	2.09	0.84	12.00	147	257	110	519	**4,862**		
42.00	(56.70)	s	50	50	22.00	40.00	-	6.70	0.36	6.85	2.45	0.84	12.00	147	301	110	519	**5,380**		

Bored Pile Ultimate Capacity Calculation

Pile diameter 800 mm

Project Road Name:	SRTPPF-2 Geotechnical Investigation, Dhaka(Katchpur)-Sylhet-Tamabil Road(N2) 283km.			Pile Diameter (m):	0.800
Bridge ID:	266+240	Borehole ID:	BH NO: 266+240	Surface Area per lm(sqm):	2.514
Bridge Loacation:	Name of place: Dhaka-Sylhet-Tamabil Road.			Pile Toe Area (sqm) :	0.503

SPT Energy correction Factor : (Trip Hammer used) 1.00	qs=Ks*Po*tan(δ)	Dc= 16	Qb= P0*Nq*Ab=	4,023

Depth, m	Red. Level	Soil type	N-value		γ	ϕ	c	α	K_0=1-sinϕ	K_s/k_0	ks	tan(δ)	γ_{sub}	p_0= γD,	Shaft Resistance ultimate				Q_b	Total ultimate
			Field	Corr'd SPT											Unit q_s	Corr q_s	qs(segment)	Q_s		Q T
	PWD			N60	kN/cum		kPa						kN/cum	kN/sqm	kPa	kPa	kN	kN	kN	kN
0	-14.70																			
1.50	(16.20)	c	4	4	14.75	FALSE	20	0.70	1.00	0.85	0.85	FALSE	4.75	7	14	14	53	**53**		
3.00	(17.70)	c	3	3	14.50	FALSE	15	0.70	1.00	0.85	0.85	FALSE	4.50	14	11	11	40	**92**		
4.50	(19.20)	c	3	3	14.50	FALSE	15	0.70	1.00	0.85	0.85	FALSE	4.50	20	11	11	40	**132**		
6.00	(20.70)	s	9	9	16.50	27.70	-	0.70	0.54	0.85	0.46	0.45	6.50	39	8	8	30	**162**		
7.50	(22.20)	s	15	15	17.75	29.50	-	0.70	0.51	0.85	0.43	0.45	7.75	58	11	11	43	**205**		
9.00	(23.70)	s	22	22	18.53	31.60	-	0.70	0.48	0.85	0.40	0.45	8.53	77	14	14	53	**257**		
10.50	(25.20)	s	18	18	18.25	30.40	-	0.70	0.49	0.85	0.42	0.45	8.25	87	16	16	62	**319**		
12.00	(26.70)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	101	19	19	72	**391**		
13.50	(28.20)	s	17	17	18.15	30.10	-	0.70	0.50	0.85	0.42	0.45	8.15	110	21	21	79	**470**		
15.00	(29.70)	s	14	14	17.52	29.20	-	0.70	0.51	0.85	0.44	0.45	7.52	113	22	22	83	**553**		
16.50	(31.20)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	113	21	21	80	**633**		
18.00	(32.70)	s	19	19	18.45	30.70	-	0.70	0.49	0.85	0.42	0.45	8.45	113	21	21	80	**713**		
19.50	(34.20)	s	14	14	17.52	29.20	-	0.70	0.51	0.85	0.44	0.45	7.52	113	22	22	83	**796**		
21.00	(35.70)	s	6	6	15.75	26.80	-	0.70	0.55	0.85	0.47	0.45	5.75	113	24	24	89	**885**		
22.50	(37.20)	s	8	8	16.25	27.40	-	0.70	0.54	0.85	0.46	0.45	6.25	113	23	23	88	**973**		
24.00	(38.70)	s	12	12	17.30	28.60	-	0.70	0.52	0.85	0.44	0.45	7.30	113	23	23	85	**1,058**		
25.50	(40.20)	s	18	18	18.25	30.40	-	0.70	0.49	0.85	0.42	0.45	8.25	113	21	21	80	**1,139**		
27.00	(41.70)	s	31	31	19.10	34.30	-	0.70	0.44	0.85	0.37	0.45	9.10	113	19	19	71	**1,210**		
28.50	(43.20)	s	28	28	18.95	33.40	-	0.70	0.45	0.85	0.38	0.45	8.95	113	19	19	73	**1,283**		
30.00	(44.70)	s	33	33	19.25	34.90	-	0.70	0.43	0.85	0.36	0.70	9.25	113	29	29	108	**1,391**		
31.50	(46.20)	s	39	39	19.49	36.70	-	0.70	0.40	0.85	0.34	0.74	9.49	113	29	29	108	**1,499**		
33.00	(47.70)	s	45	45	20.25	38.50	-	0.70	0.38	0.85	0.32	0.79	10.25	113	29	29	109	**1,608**		
34.50	(49.20)	s	47	47	21.00	39.10	-	1.70	0.37	1.85	0.68	0.81	11.00	113	63	63	236	**1,844**		
36.00	(50.70)	s	49	49	21.75	39.70	-	2.70	0.36	2.85	1.03	0.83	11.75	113	96	96	364	**2,208**		
37.50	(52.20)	s	50	50	22.00	40.00	-	3.70	0.36	3.85	1.38	0.84	12.00	113	130	110	415	**2,623**		
39.00	(53.70)	s	50	50	22.00	40.00	-	4.70	0.36	4.85	1.73	0.84	12.00	113	164	110	415	**3,037**		
40.50	(55.20)	s	50	50	22.00	40.00	-	5.70	0.36	5.85	2.09	0.84	12.00	113	198	110	415	**3,452**		
42.00	(56.70)	s	50	50	22.00	40.00	-	6.70	0.36	6.85	2.45	0.84	12.00	113	232	110	415	**3,867**		